CHAO-JEN WONG, PH.D.

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Senior bioinformatician with 15+ years of experience in genomic and epigenomic data analysis, workflow engineering, and scientific support for sequencing-based research. Expertise in PacBio long-read WGS, CUT&RUN, multi-omic NGS data analysis, Nextflow pipeline development, and integrating LLM-assisted tools into bioinformatics workflows. Experienced in partnering with wet-lab scientists and collaborators to translate experimental questions into sequencing strategies, robust QC and analysis, biomarker discovery, and biological interpretation.

TECHNICAL SKILLS

Languages: R (Package Development, Bioconductor), Python (Pandas, Scikit-Learn, TensorFlow), SQL, MATLAB, Bash

Workflows: Nextflow, Agentic workflow, Git, Docker/Singularity, AWS/EC2, HPC

Genomics: Fiber-seq, long-reads, scRNA-seq, RNA-seq, ATAC-seq, ChIP-seq, CUT&RUN, Ribo-seq, DNA methylation, variant calling, proteomics, chromatin structure

PROFESSIONAL EXPERIENCE

Sr. Bioinformatician — Oct 2023 – Present

Seattle Children's Research Institute, Seattle, WA

- Architected and extended the open-source Nextflow nf-core/pacvar (v1.1.0) pipeline—an industry-grade workflow ecosystem trusted across pharma and academia—enabling comprehensive variant calling and multi-omic methylation profiling for PacBio long-read WGS data.
- Engineered and validated an AI-assisted long-read WGS diagnostic workflow for pediatric cancers, consolidating months-long testing across 10+ molecular assays into a single rapid test that increased diagnostic yield and enabled timely precision treatment decisions.
- Uncovered the epigenetic adaptational landscape of sarcoma cells under BRD4 inhibition, identifying H3K27ac recruitment as the key mechanism triggering downstream mesenchymal shift and axonogenesis.
- Challenged a decades-old paradigm in cancer therapy by proving anthracyclines suppress aggressive leukemias through translation inhibition rather than DNA damage—a discovery enabled by integrating multi-omic and high-throughput drug screening datasets.
- Engineered R packages like *talusR* and *peakable* to automate proteomics profiling and CUT&RUN replicate validation, eliminating technical noise and preventing false-positive biological conclusions in multi-omic workflows.

Bioinformatician III — Oct 2015 – Jun 2023

Tapscott Lab, Human Biology Department, Fred Hutchinson Cancer Center, Seattle, WA

- Led data-driven research for a cross-functional initiative that discovered the primary disease progression mechanisms in facioscapulohumeral muscular dystrophy (FSHD), published in Human Molecular Genetics.
- Identified the first validated panels of gene signatures representing FSHD molecular activity, setting the industry standard for post-therapeutic clinical trial measurements.
- Pioneered research demonstrating immune cell infiltration in FSHD-affected muscle, shifting the scientific understanding of the disease's systemic progression.
- Founded and led a peer-to-peer data science community at Fred Hutch, growing membership from 32 to over 250 scientists within a year.

Bioinformatics Analyst II — Oct 2011 – Oct 2015

Grady Lab, Clinical Research Division, Fred Hutch Cancer Center, Seattle, WA

- Contributed to the first published study reporting human colon DNA methylation profiles and subtype classification in colon adenocarcinoma, using Illumina 450K array-based methylation profiling.

- Collaborated on two peer-reviewed studies on DNA methylation-based modeling of the human biological clock and epidemiologic analysis of cigarette smoking associations with prostate cancer outcomes.

Bioconductor Core Developer **Sep 2008 – Sep 2011**

Gentleman Lab, Computational Biology Program, Fred Hutchinson Cancer Center, Seattle, WA

- Contributed to the core infrastructure of Bioconductor for genomic data analysis and visualization.
- Developed novel algorithms for multi-channel flow cytometry normalization using functional data analysis.
- Served as a lead reviewer for 50+ Bioconductor package submissions, enforcing rigorous standards for statistical validity and software quality.

Postdoctoral Researcher **Aug 2007 – May 2008**

San Diego State University, San Diego, CA

- Research fields included numerical methods for solving partial differential equations and microarray analysis investigating associations between antidiabetic medications and cardiovascular disease.

Analytic Consultant **Apr 2006 – Jan 2007**

Pacific Nanotechnology (Contract), Irvine, CA

- Developed atomic force microscopy tip deconvolution and image reconstruction algorithms to correct imaging artifacts caused by imperfect probe-tip geometry, resulting in a peer-reviewed publication.

Academic Part-Time Engineer **Apr 2000 – Sep 2005**

Jet Propulsion Laboratory, Pasadena, CA

- Developed high-fidelity Telecom Forecaster Predictor (TFP) software in MATLAB to simulate deep-space communication links for NASA JPL missions, including the Mars Exploration Rovers Spirit and Opportunity.
- Co-authored volumes in JPL's Deep Space Communications and Navigation Center of Excellence (DESCANSO) book series.

EDUCATION

Ph.D. & M.S. in Mathematics **2001 – 2006**

Claremont Graduate University, Claremont, CA

Relevant coursework: statistical inference, time series analysis, and partial differential equations.
Thesis: Simulation of Immobilized Enzyme Kinetics and Transport in Sessile Hydrogen Drops. Adviser: Ali Nadim.

M.A. & B.A. in Applied Mathematics **1996 – 2000**

California State University, Fullerton, CA

RECENT PUBLICATIONS

• Sadeeshkumar H, Wong CJ, et al. Anthracyclines target the chromatin-translation axis independent of DNA damage. Manuscript under review at *Nature*.

• Wong CJ, et al. Regional and bilateral MRI and gene signatures in facioscapulohumeral dystrophy. *Human Molecular Genetics*. 2024;33(8):698-708.

• Hamm DC, Paatela EM, Bennett SR, Wong CJ, et al. The transcription factor DUX4 orchestrates translational reprogramming. *PLOS Biology*. 2023;21(9):e3002317.

• Wong CJ, et al. Canine DUXC: implications for DUX4 retrotransposition and preclinical models of FSHD. *Human Molecular Genetics*. 2022;31(10):1694-1704.

PROFESSIONAL ACTIVITIES

• Nextflow Ambassador, 2026.

• Poster presenter, Nextflow Summit Boston, 2026.

• Contributor to nf-core and participant in Nextflow Hackathons, 2024-2026.

COURSERA CERTIFICATIONS

• Sequence Models (DeepLearning.AI)

• Convolutional Neural Networks (DeepLearning.AI)

• Neural Networks and Deep Learning (DeepLearning.AI)

• Applied Machine Learning in Python (U. Michigan)

• SQL Basics for Data Science Specialization (UC Davis)